1 Basic Loiter / Turn Calc. (for Phase 4 Turn) 

- $C_{L,max}$
- Straight line stall speed: $V_{stall} = \sqrt{\frac{2(\frac{W}{S})}{\rho C_{L,max}}}$
- max Endurance speed: $V_{endurance}$ = max of $C_L^{3/2}/C_D$

I ~~max~~ Endurance speed within 20% of stall speed, if so increase with safety factor.

Safe speed ≈ 1.2 (max end. speed)

- Assume turn radius: R
- ~~fo~~ use safe speed: $V_{guess}$
- Bank angle: $\phi = \tan^{-1}(V_{guess}^2/(Rg))$
- Load factor: $n = \frac{1}{\cos\phi}$
- Turning stall speed: $V_{stall,\,Turn} = V_{stall}\sqrt{n} = \sqrt{2(W/S)n/(\rho C_{L,max})}$
- Turning safe speed: $V_{safe,\,turn} = 1.2 \cdot V_{stall,\,Turn}$
- Updated Load factor: $n = 1/\cos(\tan^{-1}(V_{safe,\,turn}^2/(Rg)))$ R = radius, g = gravity
- Turning $C_L$: ~~#~~ $C_{L,turn} = (2(W/S)n)/(\rho \cdot V_{safe,\,turn}^2)$
- Turning $C_D$: $C_{D,turn} = C_D = C_{D0} + K_1 C_L^3 + K_2 C_L$
- Turn Efficiency: $L/D = C_{L,Turn}/C_{D,turn}$
- Endurance Factor: $e = C_{L,turn}^{3/2}/C_{D,turn}$

Phase 5: Approach $\frac{W}{S_{ref}} = q \cdot C_L$ 

$L$ | Phase 4: Approach

- Approach speed safety factor

Phase 5 based on lift thus mainly wed lift force, a vertical line,

- Design point is chosen by choosing the smallest aircraft design, ~~we want~~ usually want to choose highest wing loading & lowest thrust to weight ratio

Phase 6: Horizontal Acceleration

$$\frac{T}{W} = \frac{D}{W} + \frac{a}{g_m}$$

e, endurance factor, the efficiency of a propeller-driven aircraft: $\frac{1}{g} \cdot \frac{\eta_p}{BSFC} \cdot \frac{L}{D}$

- BSFC = Brake-specific fuel consumption

- $\frac{V_2}{V_1} = \left(\frac{P_2}{P_1}\right)^{\frac{1}{3}}$